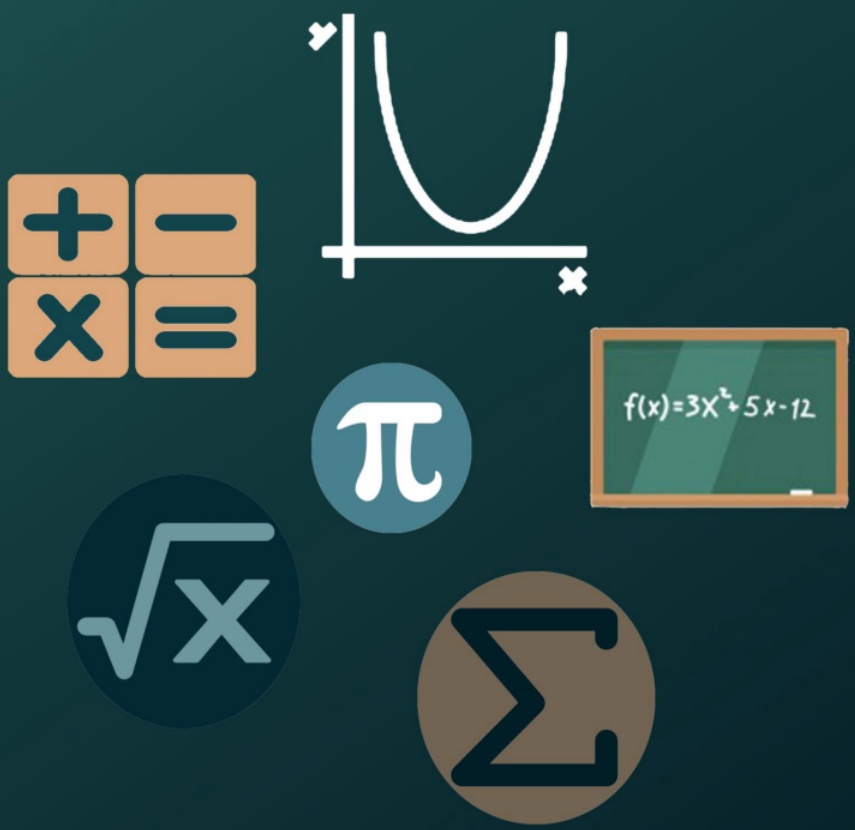
MATHEMATICS

Height & Distance



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topics	Ex-1	Ex-2
1	Height and Distance	1-9	1-9

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A bird is sitting on the top of a vertical pole 20 m high and its elevation from a point O on the ground is 45°. It flies off horizontally straight away from the point O. After one second, the elevation of the bird from O is reduced to 30°. Then the speed (in m/s) of the bird is :

(A) $AB = \frac{7\sqrt{3}}{2}(\sqrt{3}+1)$

(B) $20\sqrt{2}$

(C) $20\left(\sqrt{3}-1\right)$

(D) $40\left(\sqrt{2}-1\right)$

Q2. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle α elevation of the point A from a certain point C on the ground is 60^0. He moves away from the pole along the line BC to a point D such that CD = 7 m. From D the angle of elevation of the point A is 45^0. Then the height of the pole is :

(A) $\frac{7\sqrt{3}}{2}.\frac{1}{\sqrt{3}-1}$m (B) $\frac{7\sqrt{3}}{2}.(\sqrt{3}+1)$m

(C) $\frac{7\sqrt{3}}{2}.(\sqrt{3}-1)$m (D) $\frac{7\sqrt{3}}{2}.\frac{1}{\sqrt{3}+1}$

Q3. The length of the shadows of a veritcal pole of height h, thrown by the sun's rays at three different moments are h, 2h and 3h. The sum of the angles of elevation of the rays at these three moments is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Q4. From the bottom of a pole of height h, the angle of elevation of the top of a tower is α. The pole subtends an angle β on the top of the tower. The height of the tower is :

(A) $\frac{h\sin\alpha\sin(\alpha-\beta)}{\sin\beta}$

(B) $\frac{h\sin\alpha\cos(\alpha+\beta)}{\cos\beta}$

(C) $\frac{h\sin\alpha\cos(\alpha-\beta)}{\sin\beta}$

(D) $\frac{h\sin\alpha\sin(\alpha+\beta)}{\cos\beta}$

Q5. A house subtends a right angle at the window of an opposite house and the angle of elevation of the window from the bottom of the first house is 60°. If the distance between two houses be 6m, then the height of the first house is :

(A) $8\sqrt{3}$m (B) $6\sqrt{3}$m

(C) $4\sqrt{3}$m (D) none of these

Q6. The angle of elevation of top of a tower from a point on the ground is 30° and it is 60°, when it is viewed from a point located 40 m away from the initial point towards the tower. The height of the tower is :

(A) $20\sqrt{3}$m (B) $\frac{\sqrt{3}}{20}$m

(C) $10\sqrt{3}$m (D) $\frac{\sqrt{3}}{10}$m

Q7. The shadow of a pole of height $(\sqrt{3}+1)$ metres standing on the ground is found to be 2 metres longer when the elevation is 30° than when elevation was α. Then, α =

(A) 75° (B) 60°

(C) 45° (D) 30°

Q8. A man from the top of a 100 metres high tower sees a car moving towards the tower at an angle of depression of 30°. After some time t, the angle of depression becomes 60°. The distance (in metres) travelled by the car during this time is :

(A) $100\sqrt{3}$ (B) $\frac{200\sqrt{3}}{3}$

(C) $\frac{100\sqrt{3}}{3}$ (D) $200\sqrt{3}$

Q9. A tree 15 m high is broken by the wind at a height x metres from its bottom in such a way that top struck the ground at certain angle and horizontal distance from the root of the tree to the point where the top meet the ground is $5\sqrt{3}$ m. Determine x and find the angle of elevation made by the top of the tree with the ground.

(A) 75° (B) 30°

(C) 45° (D) 60°

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A vertical tower stands on a declivity which is inclined at 15° to the horizon. From the foot of the tower a man ascends the declivity for 80 feet and then finds that tower subtends an angle of 30°. The height of the tower is-

(A) $20(\sqrt{6}-\sqrt{2})$

(B) $40(\sqrt{6}-\sqrt{2})$

(C) $40(\sqrt{6}+\sqrt{2})$

(D) None of these

Q2. AB is a vertical pole. The point A of pole AB is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC substends an angle β at P. If AP = n AB, then tan β is equal to -

(A) $\frac{n}{2n^2+1}$ (B) $\frac{n}{n^2-1}$

(C) $\frac{n}{n^2+1}$ (D) None of these

Q3. The top of a hill observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. The height of the hill is-

(A) $\frac{h\cot q}{\cot q-\cot p}$ (B) $\frac{h\cot p}{\cot p-\cot q}$

(C) $\frac{h\tan p}{\tan p-\tan q}$ (D) None of these

Q4. A and B are two points 30 m apart in a line on the horizontal plane through the foot of a tower lying on opposite sides of the tower. If the distance of the top of the tower from A and B are 20 m and 15 m respectively, the angle of elevation of the top of the tower at A is-

(A) $\cos^{-1}(43/48)$

(B) $\sin^{-1}(43/48)$

(C) $\cos^{-1}(29/36)$

(D) $\sin^{-1}(29/36)$

Q5. A vertical pole subtends an angle $\tan^{-1}(1/2)$ at a point P on the ground. The angle subtended by the upper half of the pole at the point P is-

(A) $\tan^{-1}(1/4)$ (B) $\tan^{-1}(2/9)$

(C) $\tan^{-1}(1/8)$ (D) $\tan^{-1}(2/3)$

Q6. The angle of elevation of the top of a tower standing on a horizontal plane from a point A is α. After walking a distance d towards the foot of the tower, the angle of elevation is found to be β. The height of the tower is-

(A) $\frac{d\sin\alpha\sin\beta}{\sin(\beta-\alpha)}$ (B) $\frac{d\sin\alpha\sin\beta}{\sin(\alpha-\beta)}$

(C) $\frac{d\sin(\beta-\alpha)}{\sin\alpha\sin\beta}$ (D) $\frac{d\sin(\alpha-\beta)}{\sin\alpha\sin\beta}$

Q7. The upper $\frac{3}{4}$th portion of a vertical pole subtends an an angle $\tan^{-1}\frac{3}{5}$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is-

(A) 80 m (B) 20 m

(C) 40 m (D) 60 m

Q8. A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30°. The breadth of the river is-

(A) 20 m (B) 30 m

(C) 40 m (D) 60 m

Q9. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that AB (=a) subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30°. The height of the tower is-

(A) $2a/\sqrt{3}$ (B) $2a\sqrt{3}$

(C) $a/\sqrt{3}$ (D) $a\sqrt{3}$

Answer Key

Ex - 1								
1	2	3	4	5	6	7	8	9
C	B	A	C	A	A	C	B	B

Ex - 2								
1	2	3	4	5	6	7	8	9
B	A	B	A	B	A	C	A	C

-: Important Notes & Formulas :-